

# DO MINH DUC, Master Student

## Applying for: PHD

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### SUMMARY

- Highly motivated computer science graduate with extensive experience in Machine Learning (ML) and Deep Learning (DL).
- Proven track record of developing innovative solutions in Predictive Modeling, Reinforcement Learning, and Image Classification.
- Skilled in Data Analysis, Algorithm Development, and Quantum Computing applications.
- Experienced in Cyber Security and Android Programming.
- Passionate about advancing research in ML and DL, with a strong foundation in various programming languages and technologies.
- Eager to contribute to cutting-edge projects and further knowledge in the field of Artificial Intelligence.

### EDUCATION

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|-----------------------------------------------------------|---------------------|
| <b>Hanoi University of Science and Technology</b>         | 📍 Hanoi, Vietnam    |
| Bachelor in Computer Science                              | 📅 07/2020 - 07/2024 |
| CPA                                                       | 3.71/4.00           |
| <b>Korea Advanced Institute of Science and Technology</b> | 📍 Daejeon, Korea    |
| Master in Computer Science                                | 📅 02/2025- Now      |

### EXPERIENCE

#### Software Engineer

##### Samsung RnD Vietnam

📍 Hanoi, Vietnam 📅 06/2024 - 02/2025

##### System Security Part

- Implement and improve features of the Android Security System.
- Participate in designing, implementing, and maintaining code within the Security Team.

#### Software Engineer Intern

##### Samsung RnD Vietnam

📍 Hanoi, Vietnam 📅 03/2024 - 06/2024

##### Secure Vault

- Developed a secure vault for storing and delivering sensitive data within Android applications.
- Built a storage vault allowing applications to securely store sensitive data such as tokens and keys. Other applications can send API requests to store, request, modify, or delete data.
- Project consisted of three main modules: register, storage, and query. Developed workflows, core functionalities of the service, and managed a team of four members.
- Responsible for the register module, which allows other apps to register for the service. The app layer was built using Java, while the native layer was implemented with C++ using OpenSSL.
- Awarded Best Project by the Security Council.

#### Research Assistant

##### Korea Advanced Institute of Science and Technology

📍 Daejeon, Korea 📅 02/2025 - Present

##### VITON - Virtual Tryon

Virtual Try-On (VITON) is a specialized research area within generative modeling, particularly involving Generative Adversarial Networks (GANs) and diffusion models, aiming to realistically synthesize images of a person wearing target garments.

- Studied the full pipeline of virtual try-on systems, including human parsing (image segmentation), pose estimation, garment warping, and image synthesis.
- Investigated state-of-the-art generative approaches for clothing transfer and garment deformation modeling.
- Implemented and evaluated deep generative architectures for realistic apparel transfer while preserving human pose, body structure, and garment texture.
- Explored the integration of diffusion-based models to improve visual fidelity and structural consistency compared to traditional GAN-based approaches.

**Differential Privacy DL Optimizers Research** Researching the intersection of Differential Privacy (DP) and Deep Learning (DL), focusing on optimization algorithms that maintain privacy while preserving training stability and model performance.

- Studying privacy-preserving training methods such as DP-SGD and their limitations in large-scale deep learning.
- Investigating optimizer modifications to improve noise robustness, convergence stability, and gradient utility under DP constraints.
- Contributed to research on FIBER: A Differentially Private Optimizer with Filter-Aware Innovation Bias Correction, focusing on improving training stability under heavy noise injection.
- Analyzing optimization dynamics in DP training, including gradient filtering, delayed updates, and noise-aware adaptive learning mechanisms.

**VMMU: A Vietnamese Multitask Multimodal Understanding and Reasoning Benchmark** Research project focused on building a large-scale multimodal benchmark for Vietnamese, designed to evaluate reasoning and understanding capabilities of modern Vision-Language Models (VLMs).

- Constructed a benchmark dataset covering multiple tasks, including visual question answering, reasoning, and multimodal comprehension in Vietnamese.
- Designed evaluation protocols for testing foundation Vision-Language Models on low-resource languages.
- Developed data pipelines for multimodal annotation, quality filtering, and task diversification.
- Aimed to provide a standardized benchmark for evaluating multimodal AI systems in Vietnamese educational and reasoning contexts.

**RADBench: Evaluating Foundation VLMs for Corner Case Description on A Comprehensive Dataset**

Research project focused on evaluating foundation Vision-Language Models (VLMs) on difficult corner-case scenarios, particularly in real-world environments.

- Built a comprehensive dataset containing rare, ambiguous, and safety-critical visual scenarios.
- Evaluated multiple state-of-the-art VLMs on their ability to describe and reason about unusual visual situations.
- Proposed evaluation metrics to measure semantic correctness, contextual reasoning, and descriptive robustness.
- Aimed to improve reliability of multimodal models in safety-sensitive applications such as autonomous driving and real-world perception.

**Hanoi University of Science and Technology**

📍 Hanoi, Vietnam

📅 11/2023 - 09/2024

**Quantum Neural Network**

- Built a small QNN library that includes classes such as Feedforward, CNN, and LSTM. Researched important formulas for training networks, including gradient calculations (using the shift parameter rule for derivatives in quantum computing).
- Network structures are similar to classical neural networks, and the classes can be combined with classical networks to form hybrid networks. Used a simulator for testing due to lack of access to a quantum computer, resulting in good outcomes due to the absence of noise.
- Applied Transfer Learning in QNN to classify MNIST images, using methods similar to classical approaches.
- Proposed the Quantum Quantization Aware Training (QQAT) method to enhance the efficiency of QNNs on resource-constrained devices.

**MageCode - a code analysis system for the Bach Khoa Cyber Security Center**

- Project submitted for the Student Research Conference. Consists of three modules: analyzing whether the code was written by a human or machine, checking for code plagiarism if written by a human, and checking for logical errors such as array overflows and division by zero.
- Responsible for the third module, using LLM for error checking and CodeQL from GitHub for static analysis.
- Project is still under development at the 405th floor of the Institute of Information and Communication Technology to become a full product.
- Served as the PM, proposed the idea (with support from professors), managed the team, and designed the main workflows (determining which module the source code should enter first, which tools to use for analysis, how to conduct checks, and identifying the target audience for the product).

## Software Engineer Intern

### Viettel Telecom

📍 Hanoi, Vietnam

📅 08/2023 - 11/2023

### Viettel Enterprise Support System

- Analyzed VESS (Viettel Enterprise Support System) developed by the company in Odoo Python and produced technical documentation.
- Implemented various modules using Odoo.

## RELEVANT PROJECTS

### Graduation Thesis:

Quantum Neural Network

- **Integration Focus:** Built a quantum neural network applying transfer learning and quantization techniques.
- **Efficiency:** Explored the integration of quantum computing concepts with neural networks to enhance model efficiency and performance.

### Student Research Conference:

Code Analysis

- **Distinguishing Code:** Analyzed code to distinguish between human-written and machine-generated code using CodeBERT.
- **Vulnerability Detection:** Detected code vulnerabilities using CodeQL, contributing to software security research.
- **Similarity Identification:** Investigated the use of Dolos to identify similarities between code snippets and known code repositories or authors.

### SRV Project:

Samsung Internship Project

- **Project Focus:** Created a secure vault to securely store and deliver sensitive data in the Android environment using OpenSSL.

## SKILLS

**Operations Research:** Machine Learning, Deep Learning, Quantum Neural Network, Self - Supervised Learning | System Security | Static Analysis.

**Programming Languages:** Python, C/C++, Java.

**Tools and Frameworks:** | OpenSSL, Android Studio | PyTorch, NumPy, Pandas, Matplotlib, TensorFlow 2.0, PennyLane |  $\text{\LaTeX}$ , GitHub, Git | Jupyter Notebook.

## LANGUAGES

Vietnamese (Native)

English (Spoken & Written) : 7.5 IELTS, 930 Toeic

## ADWARDS AND HONORS

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|------------------------------------------|----------------------------------------------|
| 🏆 Prize for Student of 5 Merits          | [Hanoi University of Science and Technology] |
| 🏆 A-Level Scholarships Encouraging Study | [Hanoi University of Science and Technology] |
| 🏆 Samsung Talent Program Scholarship     | [Samsung RnD Vietnam]                        |

## PUBLICATION

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- D. D. Minh, A. T. L. Duc, D. H. Viet, N. T. Le, and Q. D. Tran, “QuantaMind: A Robust and Efficient Framework for Quantum Machine Learning Applications,” in Proc. 14th Int. Symp. Inf. Commun. Technol. (SOICT 2025), 2025.
- G. Davin, A. Vo, D. M. Do, et al., “RADBench: Evaluating Foundation VLMs for Corner Case Description on A Comprehensive Dataset,” in Proc. IEEE Intelligent Vehicles Symposium (IV), Detroit, MI, USA, Jun. 2026.
- D. M. Do, T. Do, M. S. Hoang, A. L. D. Tran, D. Kim, and H. Nguyen, “FIBER: A Differentially Private Optimizer with Filter-Aware Innovation Bias Correction,” submitted to the International Conference on Machine Learning (ICML), 2026.